

Precision Solution Dispensing for Low-Cost Quantum Dot Photodetector Fabrication

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Abstract—Lead sulfide quantum dots (PbS QDs) have shown great potential in photodetectors, solar cells, and other optoelectronic devices due to their excellent optoelectronic properties. However, traditional fabrication methods such as spin-coating suffer from low material utilization and difficulty in patterning. In this study, we propose a dispensing technology for the preparation of PbS quantum dot thin films. The effects of dispensing parameters on the optoelectronic properties of the films were systematically investigated. The results show that the dispensing technology can significantly improve material utilization, with the consumption of PbS being only 13% of that in the spin-coating method under the same coating area. Meanwhile, the prepared PbS quantum dot photodiodes exhibit good performance, with enhanced photocurrent and dark current, and shortened response time, which are comparable to those fabricated by traditional methods. This work provides a new approach for the efficient and low-cost fabrication of quantum dot optoelectronic devices.

Keywords—PbS, Quantum dot, solution dispensing, photodetector.

I. INTRODUCTION

PbS quantum dots (QDs) are promising materials for photodetection, solar cells, and light-emitting devices due to their unique optoelectronic properties such as size-tunable bandgaps and high absorption coefficients. However, the performance of devices based on PbS QDs is still limited by some factors, and the fabrication process is one of the key aspects.

Traditional fabrication processes for PbS quantum dot thin films, such as spin-coating and spray coating, have some drawbacks. For example, spin-coating requires a large amount of solution, but only a small part of the material can be effectively deposited on the substrate, resulting in low material utilization and high production cost. In addition, it is difficult to achieve patterning using these traditional methods, which limits their application in integrated optoelectronic devices. In recent years, dispensing technology has attracted attention in the field of thin film fabrication due to its advantages of precise material deposition and high material utilization. This technology can control the amount and position of the dispensed solution accurately, which is beneficial for reducing material waste and realizing patterning. Therefore, in this study, we explore the application of dispensing technology in the fabrication of PbS quantum dot thin films. The aim is to improve material utilization while ensuring the optoelectronic performance of the devices, providing a new way for the efficient fabrication of PbS quantum dot-based optoelectronic devices.

II. EXPERIMENTAL SECTION

A. Materials Preparation

PbS quantum dots were synthesized using conventional Hines method. Firstly, PbS quantum dots with large diameters were synthesized by the conventional Hines method. Next, a basic ligand exchange was performed to transfer the quantum dots from oleic acid to a mixed solution containing PbBr₂ and PbI₂ to initially passivate the quantum dot surface. Then, dual

passivation of the quantum dots was achieved by secondary ligand exchange and butylamine stabilization in lower concentrations of PbBr₂ and PbI₂ solutions, which effectively improved the surface chemical state and charge transport properties of the quantum dots. Finally, these optimally passivated quantum dots were re-dispersed and spin-coated on the electron transport layer, which was subsequently annealed to form a homogeneous and powerful photoactive layer, providing the necessary optoelectronic properties for high-performance SWIR photodetectors^[2]. The PbS quantum dot ink was prepared by dispersing the processed quantum dots in an appropriate solvent.

B. Fabrication of PbS Quantum Dot Thin Films via Dispensing Technology

The dispensing system used in this experiment consists of a precision dispensing pump, a movable stage, and a controller. The PbS quantum dot ink was loaded into a glass needle with a diameter of 50 μm. The substrate was placed on the movable stage. The fabrication process was as follows: first, the glass needle was moved to the desired position above the substrate; then, the dispensing pump was controlled to dispense a certain amount of quantum dot ink onto the substrate by adjusting the dispensing pressure and time; finally, the substrate was annealed to form a dense PbS quantum dot thin film. For comparison, PbS quantum dot thin films were also fabricated using the traditional spin-coating method under the same conditions.

C. Characterization

The morphology of the PbS quantum dot thin films was characterized by scanning electron microscopy. The current-voltage (I-V) characteristics of the photodiodes were measured using a Keithley 4200-SCS under dark and light conditions.

III. RESULTS AND DISCUSSION

A. Material Consumption

The material utilization of the dispensing technology and the traditional spin-coating method was compared by measuring the consumption of PbS quantum dot ink under the same coating area. The results show that the dispensing technology can significantly improve material utilization. The consumption of PbS in the dispensing method is only 13% of that in the spin-coating method, which is mainly due to the precise control of the ink deposition amount in the dispensing process, reducing the waste of materials.

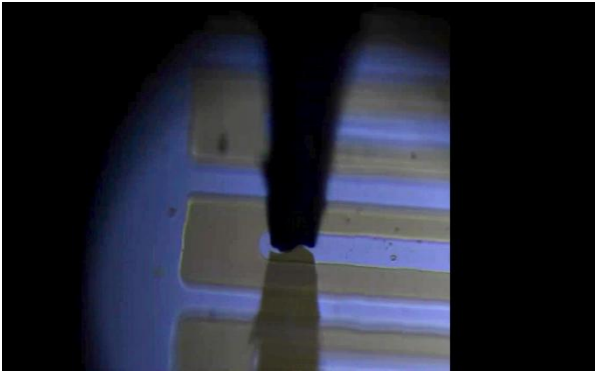


Fig.1. Microscope view of the image before dispensing

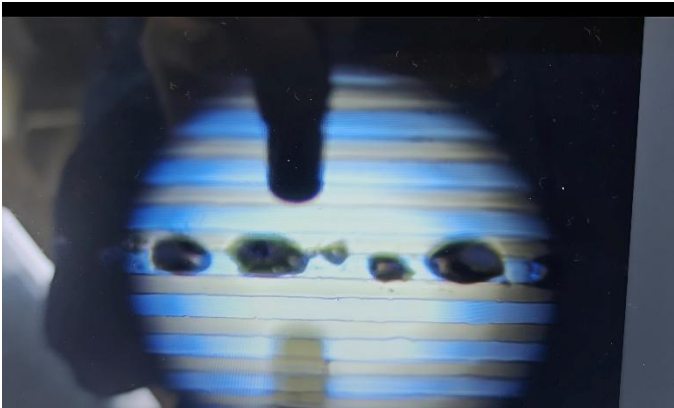


Fig.2 Microscope view of the image after dispensing

TABLE I. COMPARISON OF QUANTUM DOT COATING PROCESS PARAMETERS

Process Type	Solution Concentration (mg/ml)	Solution Volume(ul)	Quantum Dot Dosage (mg)
Dispensing method	100	4.5	0.45
Thermal spin coating method	100	40	4

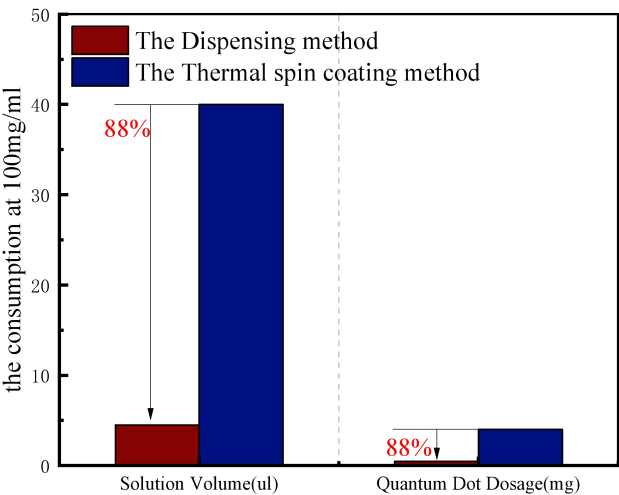


Fig.3 The Comparison of Quantum Dot Coating Process Parameters

The I-V characteristics of the PbS quantum dot photodiodes fabricated by dispensing technology were measured. The results show that the photodiodes prepared by dispensing technology exhibit enhanced photocurrent and dark current compared with some traditional devices.

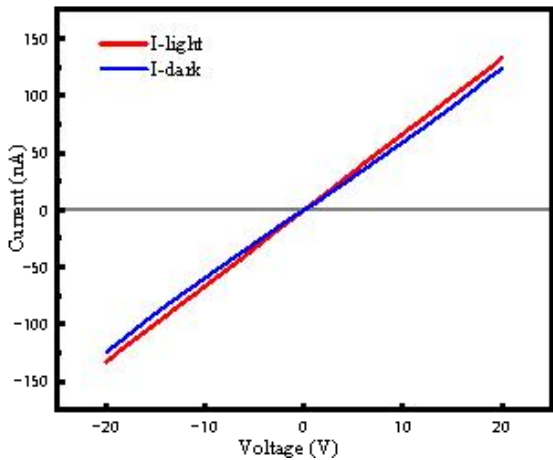


Fig.4 I - V Characteristics of Photoelectric Device (Light and Dark)

In addition, the response time of the photodiodes was tested. It is found that the dispensing technology can shorten the response time of the devices, which is important for their application in high-speed photodetection. The improved performance may be attributed to the optimized film structure and reduced defects in the films prepared by dispensing technology.

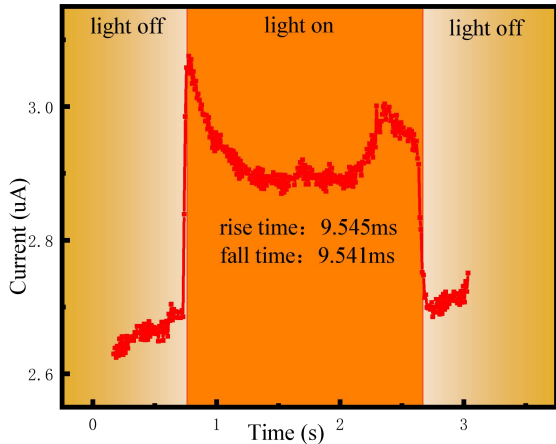


Fig.5 .Response times of photoelectric prepared by dispensing machines

Overall, the optoelectronic performance of the PbS quantum dot photodiodes fabricated by dispensing technology is comparable to that of devices fabricated by traditional spin-coating method, but with the advantage of higher material utilization.

IV. CONCLUSION

In this study, we successfully fabricated PbS quantum dot thin films using dispensing technology. The results demonstrate that compared with traditional spin-coating method, dispensing technology can significantly improve material utilization, with the PbS consumption reduced to 13% under the same coating area. The prepared thin films have uniform and dense morphology. The PbS quantum dot photodiodes based on these films show enhanced photocurrent and dark current, shortened response time, and overall performance comparable to those fabricated by traditional methods.

This work indicates that dispensing technology is a promising method for the fabrication of PbS quantum dot thin films, which can not only improve material utilization and reduce production cost but also realize the patterning of thin films, laying a foundation for the integration and application of quantum dot optoelectronic devices. Future research will focus on further optimizing the dispensing parameters to improve the performance of the devices and exploring the application of this technology in other quantum dot-based optoelectronic devices.

ACKNOWLEDGMENT

This work is supported by Shenzhen Science and Technology Program (Grant JCYJ20241202124709012,), Guangdong University Young Innovative Talents Program Project 2023 (Grant 2023KQNCX068), and Natural Science Foundation of Top Talent of SZTU (Grant GDRC202345).

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